

I B AJAY KUMAR

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EDUCATION

Ballari Institute of Technology and Management

B.E. in Computer Science and Engineering (Data Science)

CGPA: 7.75

2025 – 2029

Ballari, India

TECHNICAL SKILLS

- **Programming:** Python, C
- **Core Skills:** Problem Solving
- **Tools:** VS Code, Git

PROJECTS

1. Line Follower Robot

Technologies / Concepts: Arduino Uno, IR Sensors, L298N Motor Driver, DC Gear Motors

- Designed and developed an autonomous robot using IR sensors for real-time line detection and Arduino-based motor control.
- Implemented movement logic for forward, left, right, and stop operations based on sensor readings.
- Demonstrated practical concepts of embedded systems, sensor interfacing, motor control, automation, and basic control logic.

2. Student Attendance System

Technologies / Concepts: Digital Attendance Management Application

- Developed an application to record, manage, and monitor student attendance digitally.
- Designed functionality for maintaining student details, recording attendance, viewing attendance records, and monitoring attendance status.
- Reduced dependence on manual attendance registers through an organized digital approach to attendance management.

3. AI Multilingual Healthcare System

Technologies / Concepts: Artificial Intelligence, Conversational AI, Multilingual Interaction

- Proposed an AI-enabled healthcare application that allows users to interact with a healthcare assistant in their preferred language.
- Designed the concept to support multilingual symptom-based interaction using language detection, translation, speech-to-text, and text-to-speech technologies.
- Provides general health information, possible conditions, precautions, and guidance while clearly positioning the system as informational support rather than professional medical diagnosis.

4. Travel Budget Splitter

Technologies / Concepts: C Programming, Structures, Arrays, Functions

- Developed an application to record, organize, and manage expenses during group trips.
- Implemented expense tracking using payer name, amount, and description, followed by calculation of total trip expenditure and individual shares.
- Simplifies group expense management and demonstrates practical use of structures, arrays, functions, loops, conditional statements, and data handling.

CERTIFICATIONS

- C Language Certificate
- Python Certificate
- Data Science Hackathon – Certificate of Participation